

encing true-to-life sound. Despite the well known fact that most of us can hear frequencies between 20 Hz and 20,000 Hz on average, this additional capability of producing inaudible frequencies is not considered a surplus. Are they justified?

To an extent, yes, having a frequency response higher than 20KHz is useful to us normal humans and not just to Batman. Echolocation jokes aside, the fact is that the performance of speakers degrade as they reach the upper and lower limits, known as the cut-off frequencies. Having an extended frequency response allows for this lowering of decibels to occur outside the audible range, thus ensuring a flatter and more accurate response.

That being said, most sound that we actually play through the speakers is not at the extreme ends of our hearing limits. Of course, different grades of speaker quality do exist, but in the general audible range, a high-res driver will not necessarily be better than a good quality driver without an extended frequency response. After all, the major portion of audio clarity depends on the source file. The bit depth of the source file is the real audio resolution, and together with the sampling rate contributes to the quality. The most headphones can help with is in the noise isolation or cancellation.

Myth #4 - Tube Amps perform better than SS Amps

Anyone with a science background knows the three-terminal transistor. Amplifiers are basically specialized transistors that control the flow of a large current (Source to Drain) using a small voltage (Gate). Vacuum tubes are the relics of mankind's progress towards this technology that is now so crucial to almost every electronic product. Around the 1970's semiconductor technology started to become usable and since then the large clunky and dangerous vacuum tubes started to be replaced by their more efficient, cost-effective and reliable solid state counterparts. Yes, vacuum tubes are basically ineffective versions of solid state devices that provide more distortion, but this is exactly why some say they sound awesome!

Tube amps are said to give out rounder, warmer and more full-sounding music. This is due to the fact that the distortion from a tube amp is even-order harmonic. For example, second order harmonic means that it is the same note but an octave higher. Thus even the distortion is in tune. Not only that, the distortion is progressive with soft clipping, which means the distortion increases proportionally with the loudness, making it all the more natural sounding. This also means that you can play softly through a tube amp and it will be clear and audible as there will be proportionally